The Ambiguity Range Framework

A Diagnostic Toolkit for Operational Evaluation of Binary Classifiers

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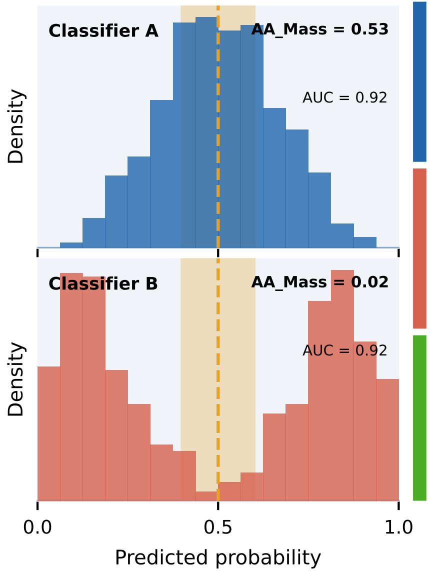
THE PROBLEM



THE FRAMEWORK

$AA_{Mass}(\delta)$

Fraction of predictions inside ℐ_δ = [0.5−δ, 0.5+δ]

Local · threshold-sensitive

AA_{PR}

Precision·recall degradation across all thresholds

Global · delta-invariant

AA_{F1}

F_β degradation across all thresholds

Global · extends to F_β

KEY FINDING

+58%

AA_Mass after Platt calibration

0.376→0.595

XGBoost raw vs calibrated

AUC flat

0.632→0.641 no discrimination gain

Theorem 2

calibration linkage formally proven

TAKEAWAYS

6

real-world datasets validated

r = 0.72

AA_Mass vs entropy non-redundant

$p < 0.05$

non-overlapping bootstrap CIs

3

formal theorems with full proofs